1.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

$$AB = \begin{matrix} a+2b & 3a+4b \\ c+2d & 3c+4d \end{matrix}$$

Each cells for AB.

① multiplication : $a,\ 2b$ $\quad N.$

② addition $\quad a+2b \quad + \quad N-1.$

$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad 2N-1.$

Multiply by number of cells in AB

$$N^2(2N-1) = 2N^3 - N^2$$

$$O(2N^3 - N^2) = O(N^3)$$

2.

Sample mean $T_n = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$

(a) $T_i = \frac{1}{n-1}\sum_{j\neq i}^{n} X_j$

$= \frac{1}{n-1}\left(\sum_{i=1}^{n} X_i - X_i\right)$

$= \frac{\sum_{i=1}^{n} X_i - X_i}{n-1}$

$= \frac{n\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) - X_i}{n-1}$

$= \frac{n\bar{X} - X_i}{n-1}$

(b) $\bar{T}_{(-1)} = \frac{1}{n}\sum_{i=1}^{n} T_{-i}$

$= \frac{1}{n}\left[\sum_{i=1}^{n}\left(\frac{n\bar{X} - X_i}{n-1}\right)\right]$ from (a)

$= \frac{1}{n}\left[\sum_{i=1}^{n}\left(\frac{n\bar{X}}{n-1}\right) - \sum_{i=1}^{n}\left(\frac{X_i}{n-1}\right)\right]$

$= \frac{1}{n}\left[\frac{n^2}{n-1}\bar{X} - \frac{1}{n-1}\sum_{i=1}^{n} X_i\right]$

$= \frac{1}{n}\cdot\frac{1}{n-1}\left[n^2\bar{X} - n\bar{X}\right]$

$= \frac{1}{n}\cdot\frac{1}{n-1}\left[n(n-1)\cdot\bar{X}\right]$

$= \bar{X}$

(c) $\widehat{Bias}_{loo} = (n-1)(\bar{T}_{(-)} - T_n)$

$= (n-1)(\bar{x} - \bar{x})$ from (b)

$= 0$

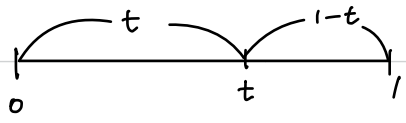
(d) $T_{loo} = T_n - 0$

$T_{loo} = T_n$

As $\widehat{bias}_{loo}$ is 0, T_{loo} is same as T_n.

3. $T_n = \min\{x_1, \cdots, x_n\}$

(a)



$P(T_i > t) = 1 - t$

$$P(T_m > t) = \prod_{i=1}^{m} P(x_i > t) = (1-t)^m$$

$$P(nT_m > t) = P(T_m > \frac{t}{n}) = (1-\frac{t}{n})^m$$

$$\log\left[(1-\frac{t}{n})^n\right] = n\log(1-\frac{t}{n})$$

Let $x = 1 - \frac{t}{n}$

$L(x) \approx f(a) + f'(a)(x-a)$

$\log(x) \approx f(1) + f'(1)(x-1)$

$\log(1) = 0 \qquad f'(x) = \frac{1}{x}$

$f'(1) = 1$

$\log(x) \approx x - 1$

$\log(1-\frac{t}{n}) \approx -\frac{t}{n}$

$n\log(1-\frac{t}{n}) \approx n \cdot -\frac{t}{n} = -t$

e^{-t}

$nT_m \xrightarrow{d} \text{Exp}(1) \qquad e^{-t}$

(b)

$T_n^* = \min \{X_1^*, \cdots X_n^*\}$

$P(T_n \text{ is not in a bootstrap sample}) = (1-\frac{1}{n})(1-\frac{1}{n}) \cdots (1-\frac{1}{n}) = (1-\frac{1}{n})^n$

$\lim_{n\to\infty} (1-\frac{1}{n})^n$ according to the definition of e^k, $e^k = \lim_{n\to\infty} (1+\frac{k}{n})^n$

$k = -1$

e^{-1}.

(c) $Pr(T_n^* = T_n)$ means that T_n is included in the bootstrap sample.

$1 - P(T_n \text{ is not included in the bootstrap sample})$

$= 1 - e^{-1}$

$= 1 - \frac{1}{e}$

$= \frac{e-1}{e}$

Sampling distribution's probability of observing $T_n = T_n$ is 0.
However, in bootstrap distribution's probability of observing
$T_n^* = T_n$ is $\frac{e-1}{e}$, which means that this nonparametric bootstrap is not valid.

4.

(a)

$Y_i(1)$ = test result when cup i has a tea first

$Y_i(0)$ = test result when cup i has a milk first

$Y_i(1) = Y_i(0)$ for all i when $i \in \{1, 2, \cdots, 8\}$

(b)



$$_8C_4 = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = 14 \times 5 = 70.$$

(c)

Get 4 pairs right

$$\binom{4}{4}\binom{4}{0} = \frac{4!\,0!}{4!} \cdot \frac{4!\,0!}{4!} = 1$$

Get 3 pairs right

$$\binom{4}{3}\binom{4}{1} = 4 \times 4 = 16$$

Get 2 pairs right

$$\binom{4}{2}\binom{4}{2} = \frac{4 \times 3}{2 \times 1} \times \frac{4 \times 3}{2 \times 1} = 36$$

Get 1 pair right

$$\binom{4}{1}\binom{4}{3} = 4 \times 4 = 16.$$

Get no pair right = 1

(d) $P = Pr(S \geq S_{obs} \mid H_0)$

S : # of correctly classified cups $\quad S_{obs} = 8$

$\frac{1}{70} = 0.0142\ldots$

(e) We can reject the null hypothesis that given the H_0 is true, the probability of having a right answer is 0.014, which is lower than threshold point 0.05.